

PDF → JSON Project — Same-Name JSON Output Guide

Purpose: This guide makes the output JSON use the same base name as the input PDF. For example: Sam.pdf → Sam.json. No date, time, timestamp, numbering, or complicated naming is used.

1. What We Want

Input PDF		Output JSON
Sam.pdf	----->	Sam.json
College.pdf	----->	College.json
Test.pdf	----->	Test.json

The important advantage is that different PDFs produce different JSON filenames. If you process Sam.pdf, the program creates Sam.json. If you later process College.pdf, it creates College.json. Sam.json is not replaced.

2. What Is Changing?

Only two things in main.py need to be changed for this requirement:

1. Tell the program which PDF file is being processed.
2. Create the JSON filename from that PDF filename.

Do not change the extraction logic. The parser, regex, STEP-ID handling, status handling, page counting, PDFInfo, and JSON structure all remain the same.

3. Current Project Structure

```
samarth/  
■■■ main.py  
■■■ pdf_parser.py  
■■■ input/  
■   ■■■ Sam.pdf  
■   ■■■ College.pdf  
■■■ output JSON files will be created in the project folder
```

For the simple version described here, the PDF you want to process is selected by changing one line in main.py.

4. Step-by-Step Code Changes

Step 1 — Open main.py

Open your current working main.py in PyCharm.

Step 2 — Add os import

At the top of main.py, you currently have:

```
import fitz
import re
import json
```

Change it to:

```
import fitz
import re
import json
import os
```

Step 3 — Find the current PDF-opening line

Currently it is:

```
pdf = fitz.open("input/input.pdf")
```

Replace it with a PDF path variable. For example, if the PDF is Sam.pdf:

```
pdf_path = "input/Sam.pdf"
pdf = fitz.open(pdf_path)
```

When you receive another PDF, only change this filename. For example:

```
pdf_path = "input/College.pdf"
pdf = fitz.open(pdf_path)
```

Step 4 — Find the current JSON-writing section

At the bottom of main.py, the current code is:

```
with open("output.json", "w") as file:
    json.dump(all_processes, file, indent=4)
```

Replace only that section with:

```
pdf_name = os.path.splitext(os.path.basename(pdf_path))[0]

output_file = pdf_name + ".json"

with open(output_file, "w") as file:
    json.dump(all_processes, file, indent=4)
```

Step 5 — Update the final print line

Replace:

```
print("JSON file created successfully.")
```

with:

```
print("JSON file created:", output_file)
```

5. Full main.py After the Change

```
import fitz
import re
import json
import os

from pdf_parser import extract_process_info, extract_status

# Select the PDF to process
pdf_path = "input/Sam.pdf"

pdf = fitz.open(pdf_path)

all_text = ""
pages_read = 0

for page in pdf:
    all_text = all_text + page.get_text() + "\n"
    pages_read = pages_read + 1

# Find every Procedure:
matches = list(
    re.finditer(
        r"Procedure\s*:",
        all_text,
        re.IGNORECASE
    )
)

procedure_blocks = []

for i in range(len(matches)):
    start = matches[i].start()

    if i + 1 < len(matches):
        end = matches[i + 1].start()
    else:
        end = len(all_text)

    block = all_text[start:end].strip()
    procedure_blocks.append(block)

# Extract information from every procedure
all_processes = []

for block in procedure_blocks:

    Procedure, Time, Dur = extract_process_info(block)

    status_data = extract_status(block)

    process_data = {
        "Procedure": Procedure,
        "Time": Time,
        "Dur": Dur,
        "status": status_data
    }

    all_processes.append(process_data)

# Add PDF information at the end
pdf_info = {
    "PDFInfo": {
        "No. of Pages read": pages_read
    }
}

all_processes.append(pdf_info)
```

```
# Create JSON with the same base name as the input PDF
pdf_name = os.path.splitext(os.path.basename(pdf_path))[0]

output_file = pdf_name + ".json"

with open(output_file, "w") as file:
    json.dump(all_processes, file, indent=4)

print("Total pages:", pages_read)
print("Total text length:", len(all_text))
print("Total procedures:", len(procedure_blocks))
print("Total dictionaries created:", len(all_processes) - 1)
print("No. of Pages read:", pages_read)
print("JSON file created:", output_file)

pdf.close()
```

6. Understanding the New Code

pdf_path stores the exact PDF being processed.

```
pdf_path = "input/Sam.pdf"
```

Python then gets only the filename without the folder and without .pdf:

```
os.path.basename(pdf_path)      → Sam.pdf
os.path.splitext(...)[0]      → Sam
```

Then .json is added:

```
output_file = pdf_name + ".json"
```

So the final output filename becomes Sam.json.

7. How to Process Another PDF

1. Put the new PDF inside the input folder.
2. Open main.py.
3. Change only the pdf_path line. Example: change "input/Sam.pdf" to "input/College.pdf".
4. Save main.py.
5. Run main.py.
6. The program creates College.json automatically.
7. The previous Sam.json remains in the project.

Example

First:
pdf_path = "input/Sam.pdf"

Result:
Sam.json

Later:
pdf_path = "input/College.pdf"

Result:
College.json

Both remain:
Sam.json
College.json

8. Very Important — What NOT to Change

- Do not change pdf_parser.py for this requirement.
- Do not change the Procedure regex.
- Do not change the Time or Dur extraction.
- Do not change the STEP-ID regex.
- Do not change Success/Failure handling.
- Do not change procedure separation.
- Do not change pages_read.
- Do not change PDFInfo.
- Do not change the JSON fields.
- Do not add datetime or timestamps.
- Do not add numbering such as output_1.json, output_2.json.
- Do not add AI, OCR, databases, or unrelated files.

9. Important Detail About Same PDF Name

If you process the **same PDF filename again**, for example Sam.pdf again, the program will create Sam.json again and therefore replace the existing Sam.json. This is intentional because the output is defined by the input filename.

For different PDF names, the outputs are different and do not replace one another:

Sam.pdf → Sam.json

College.pdf → College.json
Test.pdf → Test.json

10. Final Checklist

- ✓ Add import os.
- ✓ Create pdf_path for the PDF being processed.
- ✓ Use fitz.open(pdf_path).
- ✓ Create pdf_name from pdf_path.
- ✓ Create output_file = pdf_name + '.json'.
- ✓ Write JSON to output_file.
- ✓ Print the created filename.
- ✓ Keep all other project logic unchanged.

11. Final Expected Behavior

```
input/Sam.pdf
  ↓
  main.py
  ↓
Sam.json

input/College.pdf
  ↓
  main.py
  ↓
College.json
```

No date. No time. No timestamp. No numbering. Just the same filename with .json.